

Chandula Adhikari

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Research Interests

My primary research interest is in **computational biology**, focusing on integrating **molecular biology** with **deep learning**, **data science** and **computer vision** to develop adaptive **Human-AI collaborative systems** for healthcare diagnostics and molecular-level analysis.

Education

University of Peradeniya

BSc.Eng(Hons.) in Computer Engineering

Mar. 2021 - Aug. 2025

Current GPA: **3.85/4.0**

Dharmaraja College, Kandy

G.C.E. Advanced Level Examination - 467 from 19500+ participants nationwide

Jan. 2006 - Dec. 2019

Z-Score: **2.2069**

Publications

Optimized Multi-Processor System-on-Chip (MPSoC) Design for Low-Resource JPEG Encoding

K.H. Gunawardana, R.A.J.C. Adhikari, I. Nawinne

- Proposed a pipelined MPSoC architecture for efficient JPEG encoding using Altera Nios II/e cores on a Cyclone IV FPGA, integrating custom instructions, FIFO-based inter-core communication, and superscalar enhancements to maximize throughput.
- Presented at: ICAC 2024, Published in: IEEE Xplore

Experience

Temporary Instructor (Teaching Assistant) | Department of Computer Engineering

Sep. 2025 – Present

conducting lab sessions, preparing tutorials, lab specification sheets for courses

Intern Data Scientist | OCTAVE, Jhon Keells Holdings

July 2024 – Dec. 2024

Contributed to production-grade data pipelines, developed and validated enterprise dashboards for **Cinnamon Hotels**, and built customer segmentation solutions that directly supported digital marketing campaigns

Technologies Used: Databricks, PySpark, SQL (Spark SQL & T-SQL), Azure DevOps, MLflow, Power BI

Administrator/ Developer | *cepdnaclk*, GitHub Organization

Oct. 2023 – Present

Serve as an administrator, overseeing error resolution and repository management within the github organization of Department of Computer Engineering

Technologies Using: Jenkins, Jekyll, Fast API, GitHub

Undergraduate Teaching Assistant | Department of Computer Engineering

Nov. 2023 – July 2025

Digital Design(CO221), Computer Architecture(CO224), Third Year Project(CO300), Embedded Systems(CO321), Data Structures and Algorithms(CO1030) - Assisting labs, quizzes, and course materials, supporting students in Arm Assembly, Verilog, AVR programming and project mentoring.

Selected Projects

AI-Assisted Tool for Assessing Quality of Final Root Canal Treatment Using

Jan. 2025 – Present

Dental IOPA Radiographs | Group | 🌐

- Developing an explainable AI framework for automated root canal treatment (RCT) quality assessment using 1,000 anonymized intraoral periapical radiographs. The four-stage pipeline—comprising dataset curation, expert-guided annotation, segmentation, and multi-parameter evaluation—enables pixel- and geometry-based analysis of clinical parameters such as filling length, lateral seal, voids, and irregularities.
- Contribution:** Dataset preparation, annotation design, quality parameter evaluation, and explainable output visualization.
- Technologies:** Python, PyTorch, YOLO, OpenCV, Keras.

Analysis of RNA Velocity Reliability in Low-Dimensional Projections | Individual

Feb. 2025 – Aug. 2025

- Investigated RNA velocity reliability in low-dimensional spaces by benchmarking custom autoencoder architectures against traditional methods (PCA, t-SNE, UMAP) across 4 biological datasets, demonstrating that denoising autoencoders significantly outperform standard approaches in preserving local neighborhood structure and velocity field coherence—critical for accurate cellular differentiation trajectory inference
- Technologies:** Deep Learning (Denoising Autoencoders, VAEs, DNNs) | Dimensionality Reduction (PCA, t-SNE, UMAP) | RNA Velocity Analysis (scVelo dynamical model, Velocityto) | Evaluation Metrics (k-NN preservation, distance correlation, Jaccard similarity)

BeeZee: Smart Beehive Monitoring System | Group | |

Nov. 2023 - July 2024

- Collaborative research with the Faculty of Agriculture, University of Peradeniya to detect early signs of bee colony abscondment. Designed and built a cloud-connected data collection unit with sensors (CO_2 , humidity, temperature, weight) and camera, integrated into a real-time dashboard. Successfully implemented pollen-carrying bee detection using YOLOv8 and StrongSORT for object tracking.
- **Contribution:** Designed hardware box, assembled circuits, calibrated sensors, developed front-end dashboard, and built the computer vision pipeline for bee detection and tracking.
- **Technologies:** Raspberry Pi, AWS(S3, IoT Core, Lambda), Node.js, MongoDB, YOLOv8, StrongSORT.

CricVision: Optimizing Dynamic Batting Orders in T20 Cricket | Group | |

Mar. 2024 – Present

- Developed a data-driven system to dynamically optimize batting orders in T20 cricket using machine learning. Built a multi-output regression pipeline to predict batter performance (runs, strike rate), final team score, and Net Run Rate (NRR), enhancing decision-making after each wicket.
- **Data set :** cricsheet.org
- **Contribution:** Engineered features from Cricsheet data, web-scraped contextual data (player types, weather), implemented multiple regression models (XGBoost, Chained, Neural Networks), and deployed the best-performing model via Flask for real-time insights.
- **Technologies:** Python, Scikit-learn, XGBoost, Flask, Pandas, Matplotlib, BeautifulSoup.

Obstacle Robot Swarm for Swarm Robotic Project | Group | |

Mar. 2024 – Present

- An obstacle bot is an automated robot designed for a swarm robotic arena. It can move to desired positions, avoid collisions, and be programmed as static or dynamic obstacles.
- **Contribution:** Updated firmware to support autonomous collision handling and integrated obstacle robots into the existing swarm platform for simulating dynamic obstacle scenarios.
- **Technologies:** Arduino, Python, Java

Technical Skills

Languages	Python, C/C++, Java, JavaScript/TypeScript, Verilog HDL, ARM Assembly
Developer Tools	Git/GitHub, Unix Shell, VS Code, IntelliJ PyCharm/IDEA, Jekyll, Jenkin
Frameworks	React.js, Express.js, Flutter, Spring-Boot
Libraries	NumPy, Pandas, Matplotlib, OpenCV, TensorFlow, scikit-learn, Seaborn

Achievements / Competitions

MoraXtream 8.0 12 hour algorithmic programming competition Organized by the IEEE Student Branch of the University of Moratuwa Team Name - Five4Five, National Rank - 4(Out of 450+ participants)	Nov. 2023
IEEEExtream 17.0 24 hour algorithmic programming competition Team Name - Five4Five, Global Rank - 374(Out of 7091 participants)	Nov. 2023
Aces Coders v10.0 12 hour algorithmic programming competition Organized by the Association of Computer Engineering Students of the University of Peradeniya Team Name - Five4Five, National Rank - 12(Out of 350+ participants)	Oct. 2023
Aces PreCoders v10.0 6 hour algorithmic programming competition Team Name - Five4Five, Rank - 2(Out of 50+ participants)	Sep. 2023
NBQSA National ICT Awards IntelliSwitcher: Intelligent Domestic Energy Optimizing System 	Aug. 2023

Selected Certificates

Machine Learning Specialization Stanford University and DeepLearning.AI(Coursera) • Supervised Machine Learning: Regression and Classification • Advanced Learning Algorithms	Apr. 2024
Python for Data Science, AI and Development IBM and Coursera	Mar. 2023

Extra-Curricular Activities

Member of the Web Consultation team of University of Peradeniya	2023 – Present
Senior Member of FASE (Foundation of Astronomical Studies and Exploration)	2019 – Present
Member of Rotaract Club of University of Peradeniya	2022 – Present
Member of the Athletics team of Dharmaraja College, Kandy	2013 – 2017

References

Prof. Roshan G. Ragel | roshanr@eng.pdn.ac.lk

Professor, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka

Dr. Isuru Nawinne | isurunawinne@eng.pdn.ac.lk

Senior Lecturer, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka